# Child Vocalisation Context Classification

**Group 05** Course: Advanced Machine Learning

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All code, results, and figures are available in the project repository.

## Contents


**Abstract** **2**

**1 Introduction** **2**

**2 Dataset and Evaluation Protocol** **2**
2.1 Dataset . . . 3
2.2 Evaluation Protocol . . . 3

**3 Methods** **4**
3.1 Complete Pipeline Overview . . . 4
3.2 Audio-Level Preprocessing . . . 4
3.3 Feature Sources . . . 4
3.4 Model 1: Logistic Regression . . . 4
3.5 Model 2: Mean-Pool Transformer (Neural Baseline) . . . 5
3.6 Model 3: Attentive Transformer (Main Model) . . . 5
3.7 Training Details . . . 5
3.8 Detailed Architecture Diagrams . . . 5

**4 Hyperparameter Tuning** **8**
4.1 Model 1: Regularisation Strength — Grid Search . . . 8
4.2 Transformer Models: Architecture Search . . . 9
4.3 Extension: Bayesian Hyperparameter Search (Optuna TPE) . . . 9
4.3.1 Model 1 — Continuous C Line Search (80 trials per feature) . . . 9
4.3.2 Model 2 — Architecture + Multi-Dropout Search (60 trials per feature) . . . 9
4.3.3 Model 3 — Architecture + Multi-Dropout Search (60 trials per feature) . . . 10
4.3.4 Interpretation: Three Genuine Gains, and Val F1 as an Unreliable Selector . . . 10

**5 Experiments and Results** **11**
5.1 Main Results . . . 11
5.2 Per-class Breakdown . . . 12
5.3 The INTER Over-Prediction Problem . . . 13
5.4 Extension: Optuna Search Outcomes . . . 13
5.5 Architecture Comparison: Models 2 vs. 3 . . . 14
5.6 Error Analysis: Data-Driven Ceiling . . . 15
5.7 Regularisation Sweep . . . 15

**6 Discussion** **15**
6.1 Why Does LR Beat Transformers? . . . 15
6.2 When Did Optuna Improve a Model, and When Didn't It? . . . 15
6.3 The Language Confound . . . 16
6.4 Speaker Identity is the Dominant Bottleneck . . . 16
6.5 What Attention Concentration Tells Us . . . 16

**7 Future Work** **16**

**Conclusion** **17**

**A Supplementary Tables and Figures** **18**
A.1 Per-seed Results . . . 18
A.2 Optuna Search Parameters . . . 18
A.3 Ablation Summary . . . 19
A.4 LOCO Per-fold Results . . . 19

## Abstract

We investigate the automated classification of child vocalisation contexts from short audio segments recorded across three languages (Shipibo-Conibo, Tagalog, Vietnamese). The task is formulated as 4-class supervised classification into PLTOY, MOVE, INTER, and EXTEV categories. We compare three models: a logistic regression (LR) baseline on Mel-frequency cepstral coefficient (MFCC) summary statistics, an audio-only Transformer with mean pooling (Model 2), and a Transformer with metadata-query attentive pooling (Model 3). All models are evaluated under a participant-disjoint holdout protocol with macro-F1 as the primary metric.

The headline finding is that LR on mel spectrogram summary statistics achieves the best test macro-F1 of any model. Systematic ablations show this result is robust across three feature sources, both pooling strategies, and all hyperparameter configurations tested. Error analysis confirms that $40.9\%$ of test clips fail all three models simultaneously, indicating a data-driven ceiling. We trace the primary confound to a language mismatch between the validation and test children and quantify speaker-specific variance via Leave-One-Child-Out cross-validation ($\sigma_{\text{speaker}} \approx 0.052$ vs. $\sigma_{\text{seed}} = 0.000$ for deterministic LR).

As an extension, we apply Bayesian hyperparameter optimisation (Optuna TPE) across all 9 (model, feature) combinations. A continuous line search over $C \in [10^{-6}, 10^{6}]$ identifies $C = 0.00492$ for LR mel — falling between the discrete 9-point grid values — improving test macro-F1 from $0.277$ to $\mathbf{0.311}$, the best result of the entire project. Two further genuine gains emerge: LR wav2vec2 improves from $0.263$ to $0.285$, and Model 2 on wav2vec2 improves from $0.219$ to $0.251$ ($+0.032$ at $|\Delta|/\sigma = 1.6$). The remaining six combinations are within seed noise of canonical or degrade. The pattern correlates with the canonical val-test gap: combinations with a small gap admit further tuning, while combinations with larger validation–test mismatch are dominated by the language confound, and Optuna can worsen overfitting to the Shipibo validation child.

## Introduction

Understanding the contexts in which young children vocalise is of significant interest to developmental linguists and child language researchers. Automated methods that can classify vocalisation contexts from audio would reduce the burden of manual annotation, particularly in under-resourced language settings where annotated data are scarce.

This project addresses: *can we classify the interactional context of a child vocalisation from its acoustic content and limited metadata (age, gender)?* We work with a dataset of 1,009 clips from 9 children spanning three typologically distinct languages and four context categories.

The methodological contributions are:

- Logistic regression on mel summary statistics achieves macro-F1 $= 0.311$ (after Bayesian hyperparameter search) with 1,028 parameters, outperforming the best canonical attention transformer ($0.246$) while using far fewer parameters.
- The performance gap is not closed by regularisation, data augmentation, or larger architectures.
- A cross-linguistic confound explains the val-to-test gap: the validation and test children speak different languages with different class distributions.
- Speaker identity is the dominant variance source, dwarfing random-seed variance.
- Bayesian hyperparameter search (Optuna TPE) over a continuous search space yields three genuine gains over canonical: LR mel ($0.277 \rightarrow 0.311$), LR wav2vec2 ($0.263 \rightarrow 0.285$), and Model 2 wav2vec2 ($0.219 \rightarrow 0.251$). Most transformer searches, especially on mel, confirm that the bottleneck is the data ceiling, not hyperparameter choice.

## Dataset and Evaluation Protocol

### Dataset

The dataset consists of 1,009 short audio clips extracted from the ACQDIV corpus, annotated with one of four context categories:

| Label | Full name | Count |
|---|---|---|
| INTER | Social interaction | 325 (32.2%) |
| PLTOY | Playing with toys | 240 (23.8%) |
| EXTEV | External event | 222 (22.0%) |
| MOVE | Movement activity | 222 (22.0%) |
| Total | | 1,009 |

Clips come from 9 children spanning three languages: Shipibo-Conibo (South America), Tagalog (Philippines), and Vietnamese (Southeast Asia).

### Evaluation Protocol

With 9 participants, the problem setting specifies a **participant-disjoint fixed holdout**:

| Split | Children | Clips | Language |
|---|---|---|---|
| Train | 7 | 569 | Shipibo, Tagalog, Vietnamese |
| Validation | 1 (child_2) | 171 | Shipibo |
| Test | 1 (child_6) | 269 | Tagalog |

The validation and test children speak *different languages* — this cross-linguistic mismatch is the central confound discussed in Section 6. The primary metric is **macro-F1**. Final test scores are reported as mean $\pm$ std across 5 random seeds (42, 123, 456, 789, 2024). Seeds are used for variance estimation only, not model selection. Splits are fixed once and never rebuilt.

## Methods

### Complete Pipeline Overview

Figure 1 summarises the full pipeline. Importantly, **all three feature sources (MFCC, mel, wav2vec2) are evaluated with all three models**. The key distinction is the downstream representation: Model 1 consumes fixed-length summary statistics, whereas Models 2–3 consume padded frame sequences; metadata is used only by Model 3.

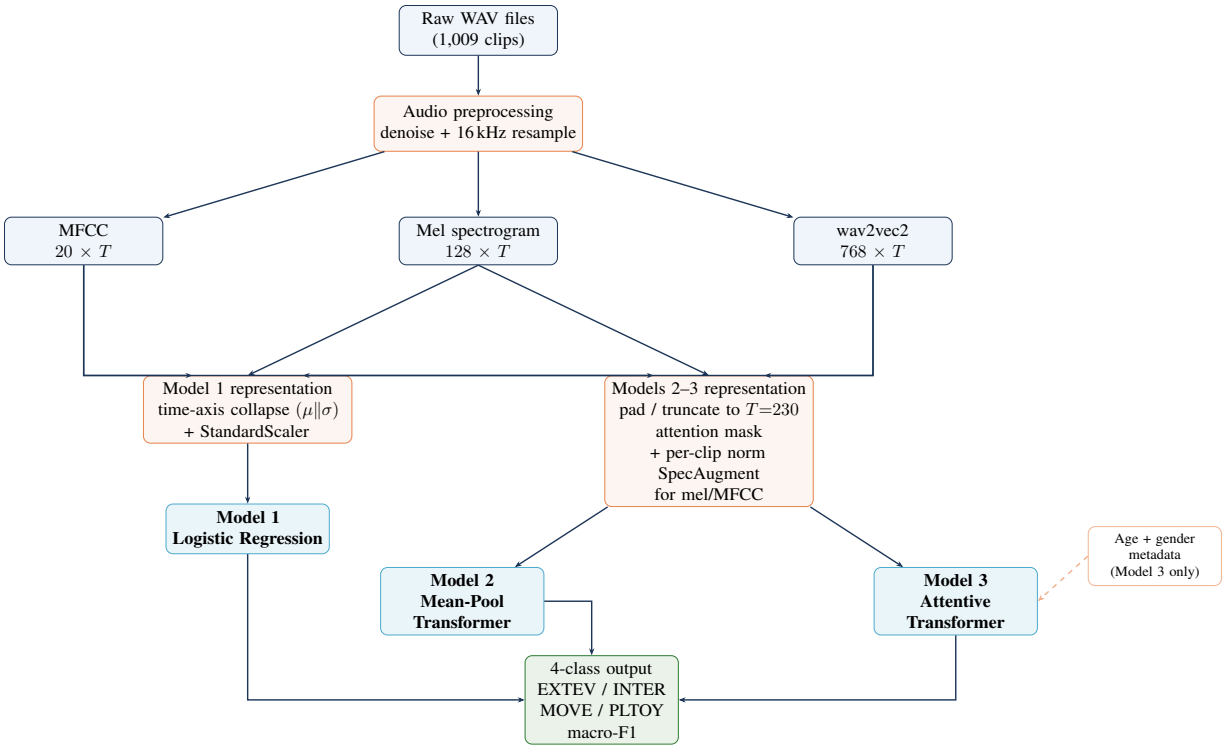



Figure 1: Complete pipeline from raw audio to classification output. All three feature sources (MFCC, mel spectrograms, wav2vec2 embeddings) are evaluated with all three models. Model 1 uses collapsed summary statistics; Models 2–3 use padded sequences, and only Model 3 additionally receives metadata.

### Audio-Level Preprocessing

All models share the same preprocessed audio. Raw recordings are first denoised using a spectral noise-reduction procedure applied once and cached. At feature extraction time, each WAV is loaded at 16 kHz using librosa.

### Feature Sources

**MFCC.** 20 Mel-frequency cepstral coefficients extracted via librosa (`n_fft`=2048, `hop_length`=512, 16 kHz). For Model 1, collapsed to a 40-dimensional vector (mean and std per coefficient). For Models 2–3, the padded $20 \times 230$ sequence is used.

**Mel spectrogram.** 128-bin log-mel spectrogram (`n_fft`=2048, `hop_length`=512, `f_min`=0 Hz, `f_max`=8 kHz, 16 kHz), cached as PNG after denoising. For Model 1, collapses to a 256-dimensional vector (mean and std per bin). For Models 2–3, the padded $128 \times 230$ sequence is used.

**wav2vec2.** Last-layer embeddings from a frozen, pretrained `wav2vec2-base` encoder (768 dimensions per frame).

### Model 1: Logistic Regression

The logistic regression baseline receives a fixed-length feature vector and applies: (1) time-axis collapse (mean $\pm$ std), (2) `StandardScaler`, (3) `class_weight='balanced'`, and (4) L2 regularisation strength $C$

tuned via inner Leave-One-Participant-Out cross-validation. Total parameters: 1,028 (mel), 164 (MFCC), or 6,148 (wav2vec2).

### Model 2: Mean-Pool Transformer (Neural Baseline)

An audio-only Transformer encoder followed by masked mean pooling. Each input sequence is per-clip normalised (z-score: subtract per-clip mean and divide by per-clip std across all time–frequency bins) before being fed to the model. SpecAugment [2] is applied during training to spectral sequence inputs (mel spectrograms and MFCCs), with two feature-axis masks ($F_{\max}$=27 bins) and two time masks ($T_{\max}$=70 frames) applied independently per clip. It is not applied to wav2vec2 embeddings. Four independent dropout locations in v2: `dropout_encoder` (inside layers), `dropout_input` (after input projection), `dropout_head` (after pooling).

### Model 3: Attentive Transformer (Main Model)

Model 3 replaces mean pooling with metadata-query attentive pooling. The metadata vector (standardised age + one-hot gender) is projected to a query vector $\mathbf{q} \in \mathbb{R}^d$. Attention weights:

$$\boldsymbol{\alpha} = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{H}}{\sqrt{d} \cdot \tau}\right), \qquad \hat{\mathbf{h}} = \mathbf{H}\boldsymbol{\alpha}^\top$$

In v2 a fourth dropout location `dropout_meta` is applied to the metadata query after projection.

### Training Details

Both transformer models (Models 2 and 3) are trained with the **Adam** optimiser and **cross-entropy loss** with **class weights** set inversely proportional to class frequency (`class_weight='balanced'`), matching the LR baseline. Batch size is **32**. Training runs for up to **100 epochs** with **early stopping** (patience = 15 on validation macro-F1); the best checkpoint by validation F1 is restored before test evaluation. The learning rate is fixed per experiment (no schedule); canonical values are $10^{-3}$ for mel/MFCC and $5\times10^{-4}$ for wav2vec2. Model 1 (LR) uses the scikit-learn `lbfgs` solver, which converges to a deterministic solution with no stochastic elements.

### Detailed Architecture Diagrams

Figures 2–4 show the complete computational graph of each model with exact tensor shapes at each stage. **Pink** nodes are new in v2: four independent dropout locations that replace the original single `dropout` parameter.

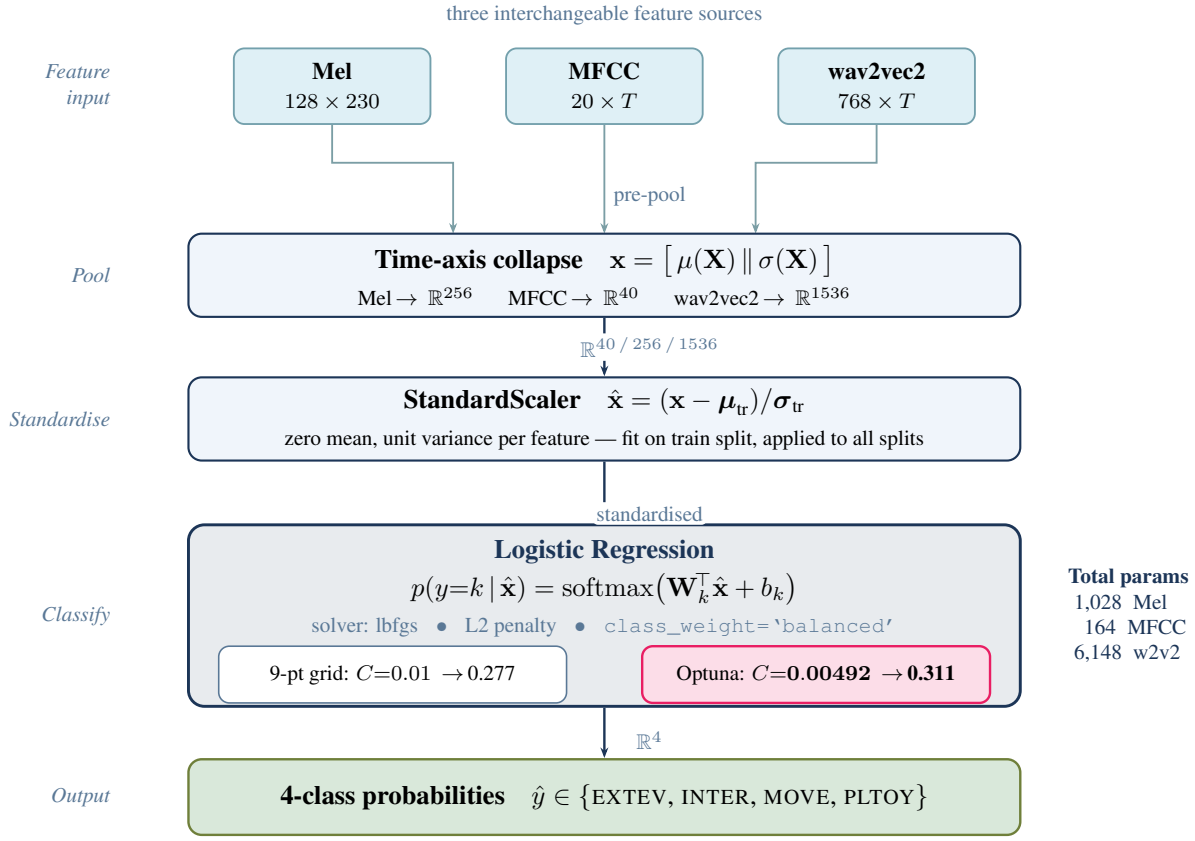



Figure 2: Model 1 (Logistic Regression) architecture. Three feature sources are first collapsed along the time axis via mean ‖ std concatenation, then standardised and passed to a class-balanced L2-regularised classifier. The model is deterministic ($\sigma$=0); per-seed variance is zero. The Optuna continuous $C$ line search (pink pill) identified $C$=0.00492 between discrete grid values, improving mel test macro-F1 from 0.277 to **0.311** — the best result of the entire project.

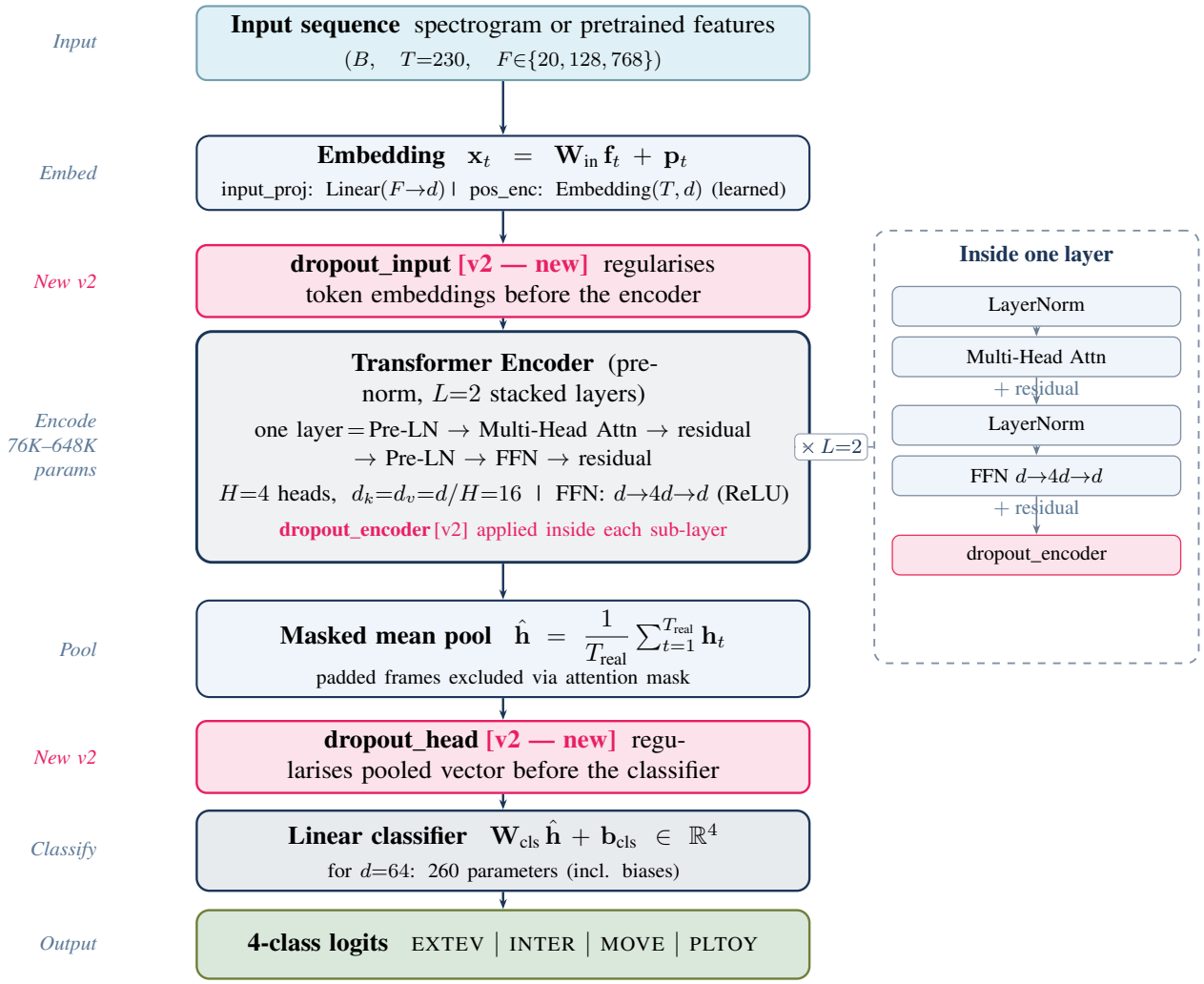



Figure 3: Model 2 (AUDIOTRANSFORMER, mean pooling) v2 architecture. The main computational graph (centre) is annotated with tensor shapes at each stage; the dashed inset (right) reveals the internal sub-layers of one Transformer encoder block, stacked $L$=2 times. Pink boxes are the three new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_head`) that replace the original single `dropout` parameter. Canonical mel config: $d$=64, $L$=2, `dropout_encoder`=0.5, lr=$10^{-3}$ (test F1: $0.249 \pm 0.020$).

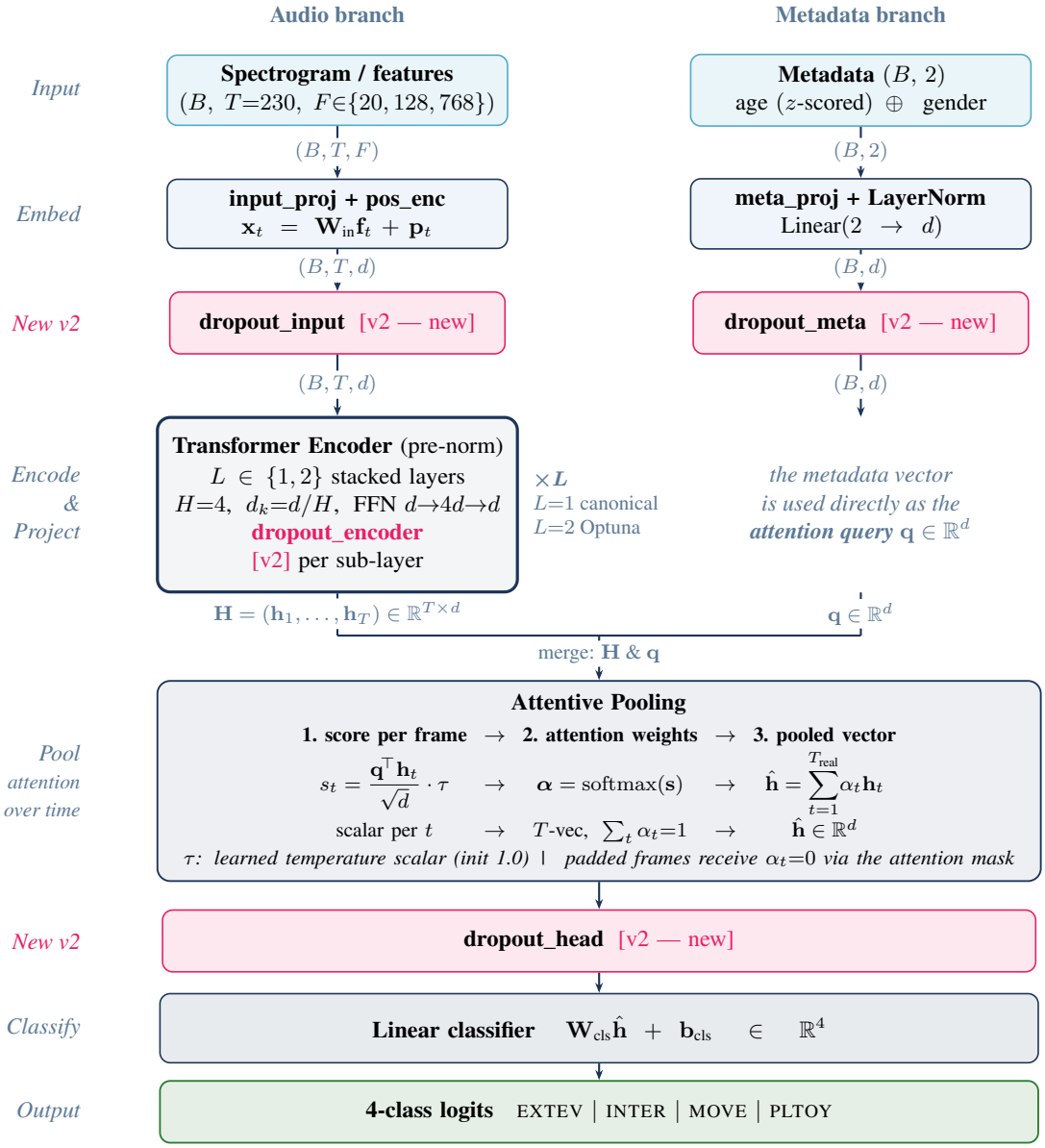



Figure 4: Model 3 (AttentivePoolingTransformer) v2 architecture. The audio branch (left) produces a sequence of contextualised hidden states $\mathbf{H}$; the metadata branch (right) produces a single vector $\mathbf{q}$ that serves as the attention query. Their merge happens inside the **Attentive Pooling** block, which scores every frame against the metadata query, softmaxes into attention weights $\boldsymbol{\alpha}$, and returns the weighted sum $\hat{\mathbf{h}}$. Pink boxes mark the four new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_meta`, `dropout_head`); `dropout_meta` is unique to Model 3. Canonical mel config: $d$=32, $L$=1, `dropout_encoder`=0.3 (test F1 $0.221 \pm 0.013$). Optuna config: $d$=64, $L$=2, lr=$1.24\times10^{-4}$ (test F1 $0.233 \pm 0.014$).

## Hyperparameter Tuning

### Model 1: Regularisation Strength — Grid Search

The L2 regularisation strength $C$ is selected via inner LOPO cross-validation over the 7 training children. The 9-point logspace grid:

$$C \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$$

| Feature | Selected $C$ | Inner-CV F1 | Test F1 |
|---|---|---|---|
| MFCC | 10 | 0.243 | 0.253 |
| Mel | 0.01 | 0.210 | 0.277 |
| wav2vec2 | 0.01 | 0.230 | 0.263 |

### Transformer Models: Architecture Search

An **8-configuration grid search** per (model, feature source) tunes $d_{\text{model}}$, `num_layers`, `dropout`, and `lr` over the following values:

$d \in \{32, 64\}, \quad L \in \{1, 2\}, \quad$ `dropout` $\in \{0.3, 0.5\}$ (lr fixed at $10^{-3}$ for mel/MFCC, $5{\times}10^{-4}$ for wav2vec2)

giving $2 \times 2 \times 2 = 8$ configurations. For Models 2 and 3 on **MFCC and wav2vec2**, the mel-optimised config is reused directly ($d$=64, $L$=2 for M2; $d$=32, $L$=1 for M3) rather than running a separate search, as the feature dimensionality differences are handled by the shared `input_proj` layer. The wav2vec2 config for M3 was manually adjusted to $d$=192 after the numerical top-1 showed artefacts from a lucky initialisation (see footnote below).

The three best-performing configurations reported in the table below are the training configurations used for all canonical results:

| Model | Feature | $d$ | L | drop | lr | Val F1 |
|---|---|---|---|---|---|---|
| Model 2 | Mel | 64 | 2 | 0.5 | $10^{-3}$ | 0.306 |
| Model 3 | Mel | 32 | 1 | 0.3 | $10^{-3}$ | 0.286 |
| Model 3 | wav2vec2 | 192 | 1 | 0.3 | $5{\times}10^{-4}$ | 0.297$^{\dagger}$ |

† Manually selected; numerical top-1 showed lucky initialisation.

### Extension: Bayesian Hyperparameter Search (Optuna TPE)

As an extension we replace the discrete grids with Bayesian optimisation (Optuna Tree-structured Parzen Estimator) across all 9 (model, feature) combinations: 80-trial line searches for LR on each of MFCC, mel, and wav2vec2; 60-trial multi-dimensional searches for M2 and M3 on each of MFCC, mel, and wav2vec2. The transformer searches additionally exercise four independent dropout locations introduced in v2 of the architecture: `dropout_encoder`, `dropout_input`, `dropout_head`, and `dropout_meta` (Model 3 only).

#### Model 1 — Continuous C Line Search (80 trials per feature)

The discrete grid is replaced by a log-uniform search over $C \in [10^{-6}, 10^{6}]$, seeded with the 9 canonical grid points, and run independently for each of the three feature sources. Optuna identifies $C = 0.00492$ for mel (between the grid values 0.001 and 0.01), $C = 0.01498$ for wav2vec2, and $C = 0.01735$ for MFCC.

| Feature | 9-pt grid Test F1 | Optuna Test F1 | Δ | Verdict |
|---|---|---|---|---|
| Mel | 0.277 | **0.311** | +0.034 | **Genuine gain** |
| wav2vec2 | 0.263 | **0.285** | +0.022 | **Genuine gain** |
| MFCC | 0.253 | 0.239 | −0.014 | Genuine loss |

Since LR is deterministic (lbfgs solver), no seed averaging is required. Mel yields the largest single gain in the project (+0.034) and establishes **0.311** as the new best result. wav2vec2 also benefits from the finer search (+0.022). MFCC shows the opposite pattern: the inner-CV F1 maximum (0.246 at $C$=0.01735, beating the grid winner 0.243 at $C$=10) did not transfer to test performance — a small-scale instance of the "val F1 as unreliable selector" phenomenon documented for the transformers below.

#### Model 2 — Architecture + Multi-Dropout Search (60 trials per feature)

The search explores $d_{\text{model}} \in \{32, 64, 128, 192\}$, $L \in \{1, 2, 3\}$, lr log-uniformly over $[10^{-4}, 10^{-2}]$, and three continuous dropout parameters. The Optuna-selected configurations are reported from 5-seed reruns,

including the corrected multiseed M2 mel rerun.

| Feature | Best ($d$, $L$, lr) | Best trial val/test | Reported test | Canonical | Δ | Verdict |
|---|---|---|---|---|---|---|
| wav2vec2 | (128, 1, $8.0\times10^{-4}$) | 0.341 / 0.260 | **$0.251 \pm 0.020$** | $0.219 \pm 0.018$ | $+0.032$ | **Genuine gain** |
| MFCC | (128, 1, $5.1\times10^{-3}$) | 0.394 / 0.216 | $0.219 \pm 0.029$ | $0.222 \pm 0.033$ | $-0.003$ | Within seed noise |
| Mel | (32, 2, $1.83\times10^{-3}$) | 0.336 / 0.207 | $0.226 \pm 0.018$ | $0.249 \pm 0.020$ | $-0.023$ | Within seed noise |

**wav2vec2 (genuine gain).** Optuna finds a wider, single-layer architecture with low input/head dropout that improves test F1 by $+0.032$ at $|\Delta|/\sigma = 1.6$. This is the only transformer configuration in the project to genuinely beat its canonical baseline. The canonical val-test gap on wav2vec2 was already small (0.057), leaving room for tuning to add signal rather than overfit.

**Mel (within-noise degradation).** The single best Optuna trial had val/test $0.336/0.207$, but the corrected 5-seed rerun gives $0.226 \pm 0.018$ on test. Compared with the canonical $0.249 \pm 0.020$, this is a drop of $\Delta = -0.023$ and $|\Delta|/\sigma \approx 1.2$, so it is a downward trend but not beyond the seed-noise threshold. Optuna does not improve M2 mel; the canonical M2 mel result ($0.249$) stands.

**MFCC (within noise).** The canonical M2 MFCC run ($0.222 \pm 0.033$) confirms the Optuna search adds no signal: the Optuna winner lands at $0.219 \pm 0.029$, a delta of $-0.003$ at $|\Delta|/\sigma < 0.1$. The val-test gap of the best Optuna trial (0.394 / 0.216) is by far the largest of any M2 run, indicating heavy overfitting to the Shipibo validation child on this feature source.

### Model 3 — Architecture + Multi-Dropout Search (60 trials per feature)

The search adds `dropout_meta` on top of M2's space and uses `MetadataSpectrogramDataset` with proper metadata standardisation.

| Feature | Best ($d$, $L$, lr) | Best trial val/test | Reported test | Canonical | Δ | Verdict |
|---|---|---|---|---|---|---|
| Mel | (64, 2, $1.24\times10^{-4}$) | 0.322 / 0.242 | $0.233 \pm 0.014$ | $0.221 \pm 0.013$ | $+0.012$ | Within seed noise |
| wav2vec2 | (64, 1, $1.69\times10^{-3}$) | 0.332 / 0.259 | $0.241 \pm 0.021$ | $0.246 \pm 0.016$ | $-0.005$ | Within seed noise |
| MFCC | (32, 2, $5.2\times10^{-4}$) | 0.324 / 0.211 | $0.217 \pm 0.038$ | $0.208 \pm 0.040$ | $+0.009$ | Within seed noise |

All three M3 Optuna 5-seed results sit within seed noise of their canonical counterparts ($|\Delta|/\sigma < 0.3$ in all cases). Per-seed values for the mel run:

| Seed | 42 | 123 | 456 | 789 | 2024 | Mean | Std |
|---|---|---|---|---|---|---|---|
| Val F1 | 0.263 | 0.263 | 0.241 | 0.232 | 0.243 | 0.248 | — |
| Test F1 | 0.246 | 0.246 | 0.239 | 0.210 | 0.222 | **0.233** | 0.014 |

The marginal mel improvement ($+0.012$) is within the seed-variance noise of $\sigma = 0.013$ but notably reduces the val-test gap from $\sim$0.065 to $\sim$0.016, confirming the lower lr reduces overfitting to the validation child.

### Interpretation: Three Genuine Gains, and Val F1 as an Unreliable Selector

Across nine search runs, Optuna achieves three genuine gains: LR mel ($+0.034$), LR wav2vec2 ($+0.022$), and M2 wav2vec2 ($+0.032$). The remaining six are within seed noise of canonical or degrade. The pattern shows that Optuna helps when the validation objective transfers to the test child: LR is deterministic and benefits from finer continuous $C$ values, while M2 wav2vec2 contains enough transferable signal for architecture search to help. When validation–test mismatch is large, especially for M2 mel, Optuna fails to improve transfer and can trend downward on the Tagalog test child.

The transformer-on-mel evidence reinforces this directly: across 60 trials each for M2 and M3, higher validation F1 does not predict higher test F1 (Figure 8). The Shipibo validation child and the Tagalog test child have different class distributions, so a model tuned to fit the Shipibo prior generalises worse to Tagalog — a finding that is structural and cannot be solved by hyperparameter search alone (see Section 6).

## Experiments and Results

### Main Results

Table 1: Test macro-F1 for all model–feature combinations. Multi-seed rows are shown as mean ± std over 5 seeds; LR is deterministic (std = 0.000). Optuna gains over canonical are shown in **pink**; genuine losses in gray; within-noise changes are unbolded.

| Model | MFCC | Mel | wav2vec2 |
|---|---|---|---|
| Model 1 (LR) canonical (9-pt grid) | **0.253** ± 0.000 | 0.277 ± 0.000 | 0.263 ± 0.000 |
| Model 1 (LR) + Optuna | 0.239 ± 0.000 | **0.311** ± 0.000 | **0.285** ± 0.000 |
| Model 2 (mean pool) canonical | 0.222 ± 0.033 | 0.249 ± 0.020 | 0.219 ± 0.018 |
| Model 2 (mean pool) + Optuna | 0.219 ± 0.029 | 0.226 ± 0.018 | **0.251** ± 0.020 |
| Model 3 (attention) canonical | 0.208 ± 0.040 | 0.221 ± 0.013 | **0.246** ± 0.016 |
| Model 3 (attention) + Optuna | 0.217 ± 0.038 | 0.233 ± 0.014 | 0.241 ± 0.021 |
| Chance (uniform) | | 0.245 ± 0.027 | |

**Key findings:**

- The Optuna line search for LR mel yields $\mathbf{0.311}$, the best result of the project — a 1,028-parameter linear model outperforms a 333K-parameter mean-pool transformer by 24% and a 117K-parameter attention transformer by 26%.
- Model 2 wav2vec2 + Optuna ($0.251 \pm 0.020$) is the **only transformer configuration to genuinely improve over its canonical baseline** ($+0.032$ at $|\Delta|/\sigma = 1.6$), and is one of the three genuine Optuna gains in the project.
- LR + Optuna improves on every feature where the canonical gap was already small (mel, wav2vec2) and degrades on MFCC, where the inner-CV maximum did not transfer to test.
- M3 mel and M3 wav2vec2 + Optuna are within seed noise of canonical; M2 mel + Optuna trends downward by $0.023$ but remains within seed noise; the corrected 5-seed val–test gap is about $0.065$.

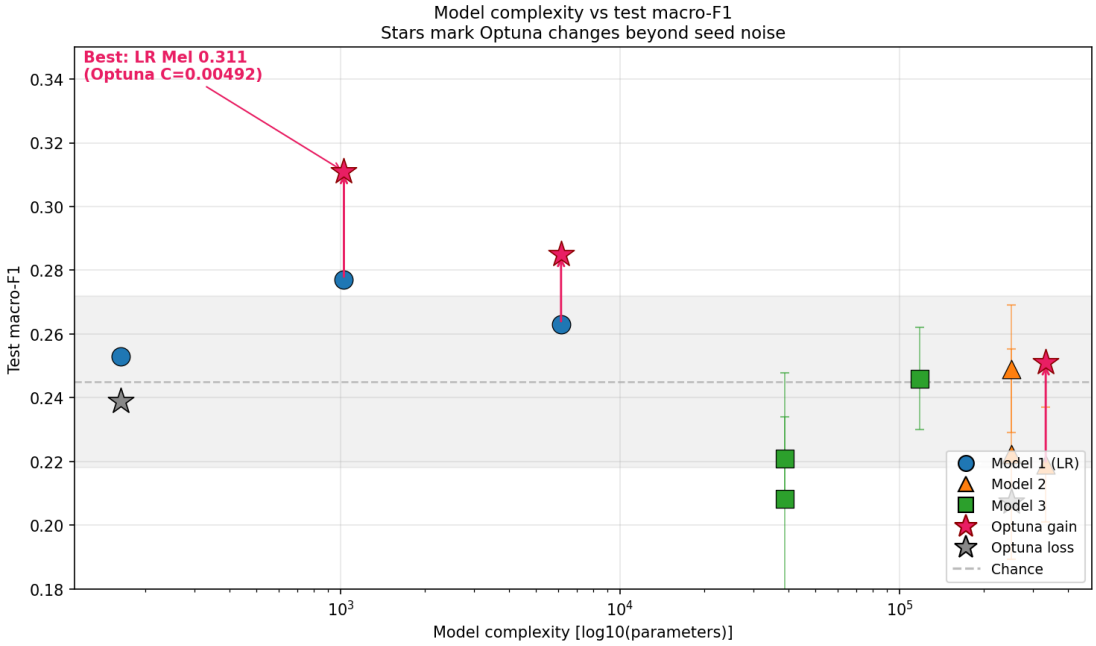



Figure 5: Test macro-F1 vs. model complexity (log scale). Stars ($\star$) show Optuna changes beyond the noise threshold; pink stars mark genuine gains and gray stars mark genuine losses. LR mel reaches $\mathbf{0.311}$ (pink star, top); LR wav2vec2 reaches $0.285$; and M2 wav2vec2 + Optuna reaches $0.251$ (pink star, mid-right). More parameters do not improve accuracy.

## Per-class Breakdown

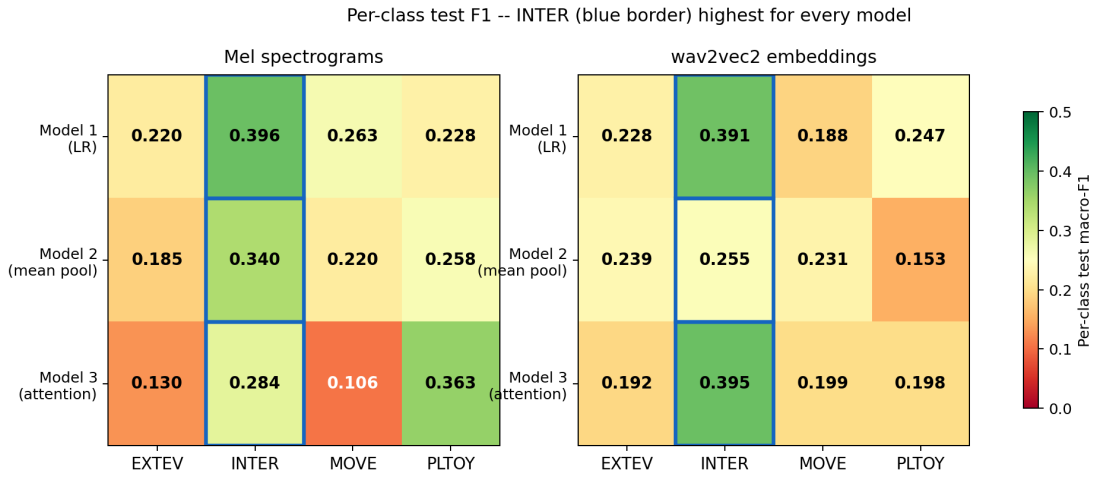



Figure 6: Per-class test F1 heatmap. INTER (blue border) is the highest-F1 class in five of six panels; the exception is Model 3 on mel, where PLTOY is highest.

The pattern is strikingly uniform across architectures: INTER is the highest-F1 class in five of six panels, with Model 3 on mel as the exception (PLTOY is highest there). EXTEV and MOVE are generally lower and unstable across models. The similarity between a 1,028-parameter LR and larger transformer baselines is strong evidence that the ceiling is *data-driven*.

## The INTER Over-Prediction Problem

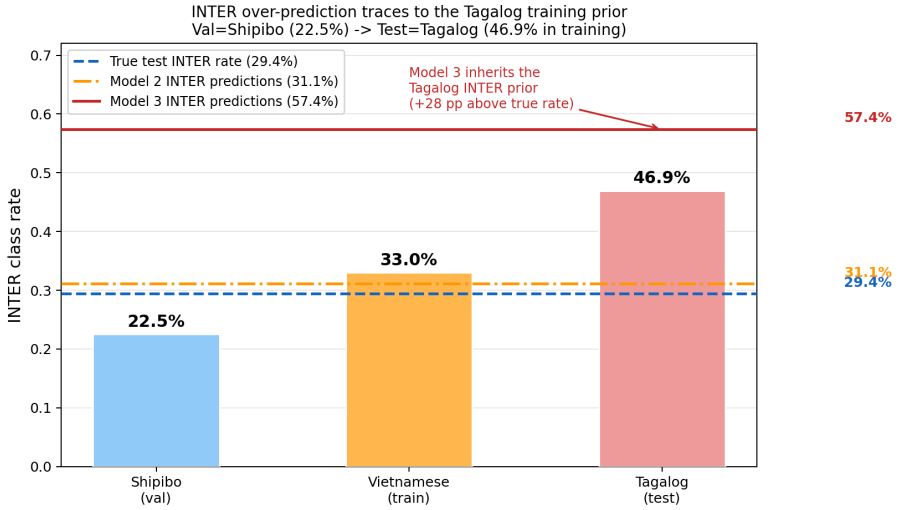



Figure 7: INTER class rate by language in training data (bars) and model prediction rates (horizontal lines). Model 3 predicts INTER on 57.4% of test clips vs. the true rate of 29.4%.

Two compounding effects drive the INTER bias: (1) the Tagalog training prior (46.9% INTER) is substantially higher than the true test rate (29.4%); and (2) Model 3 on wav2vec2 attends to a single frame in 95% of clips, which most likely carries interaction-like content.

## Extension: Optuna Search Outcomes

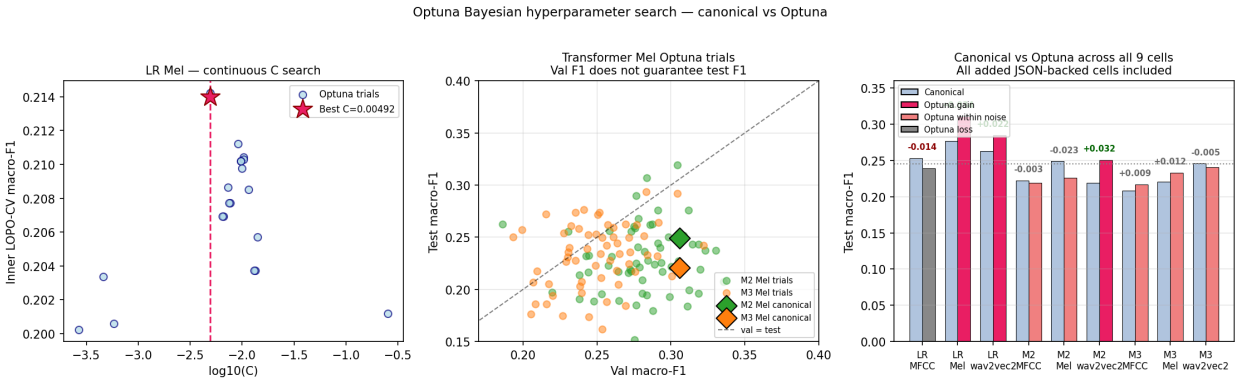



Figure 8: **Left**: LR mel continuous $C$ search — best $C$=0.00492 lies between discrete grid points. **Centre**: M2/M3 mel Optuna trials — val F1 does not predict test F1 (language confound). **Right**: canonical vs. Optuna for all 9 (model, feature) cells. Three genuine gains appear in pink: LR mel (+0.034), LR wav2vec2 (+0.022), and M2 wav2vec2 (+0.032).

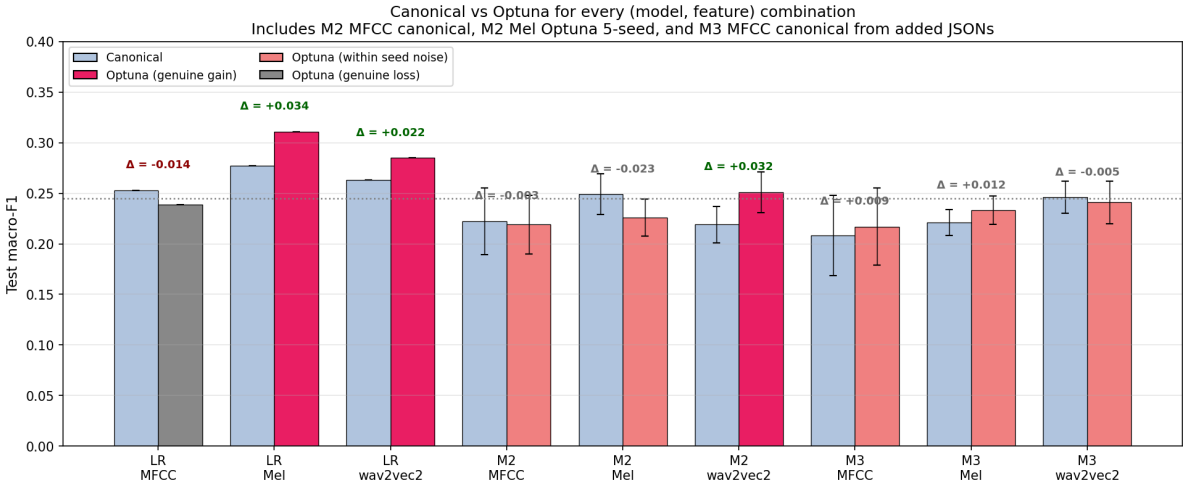



Figure 9: Canonical vs. Optuna for every (model, feature) cell, with Δ annotated above each pair. Three cells show genuine gains: LR mel (+0.034, deterministic), LR wav2vec2 (+0.022, deterministic), and M2 wav2vec2 (+0.032 at $|\Delta|/\sigma$=1.6). LR MFCC is the only genuine loss; M2 mel trends downward but remains within seed noise, as do the remaining non-gain transformer cells.

The 9-cell view in Figure 9 consolidates the extension's findings. Optuna's success splits along a clear axis: it adds signal where the validation signal transfers to the test child, and it overfits the validation child where the split mismatch is large. For deterministic LR, the mel and wav2vec2 improvements are not seed artifacts; for transformers, only M2 wav2vec2 genuinely improves because its canonical configuration was already well calibrated. M2 mel trends downward within seed noise, while the M3 mel/wav2vec2 configurations show within-noise changes, consistent with the language-confound interpretation in Section 6.

### Architecture Comparison: Models 2 vs. 3

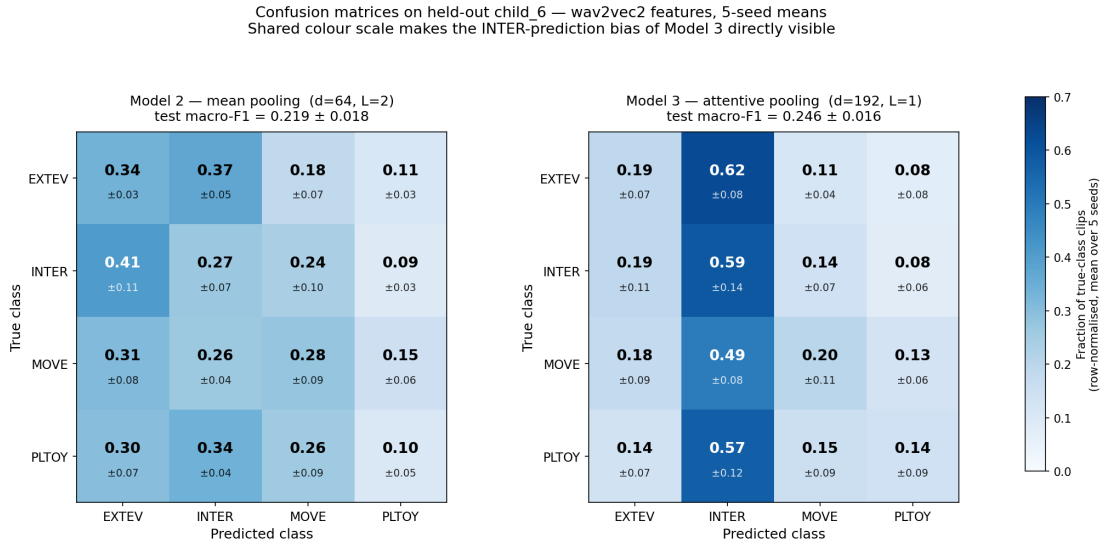



Figure 10: Row-normalised confusion matrices for Model 2 (left) and Model 3 (right) on wav2vec2, 5-seed means. Model 3 predicts INTER 57.4% of the time vs. the true 29.4%.

### Error Analysis: Data-Driven Ceiling

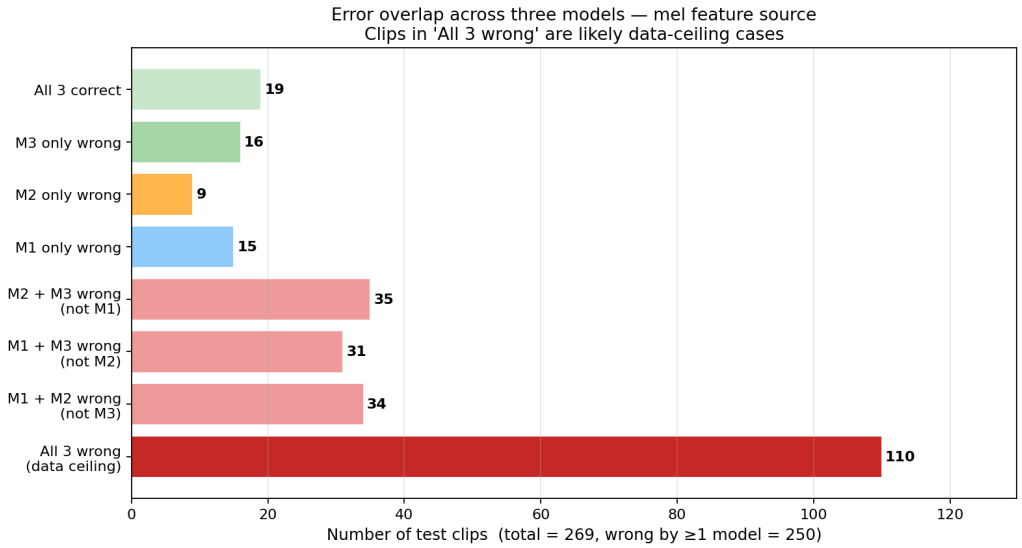



Figure 11: Error overlap: 110 clips (40.9%) are wrong by all three models simultaneously; only 19 (7.1%) are correct by all three.

The three models have nearly identical per-clip error rates (M1: 70.6%, M2: 69.9%, M3: 71.4%) and the same 110 clips fail all three architectures.

### Regularisation Sweep

A 6-configuration sweep on mel features varying dropout (0.3–0.7), weight decay ($10^{-4}$, $5\times10^{-4}$), and label smoothing (0.0–0.2) achieved $0.240 \pm 0.028$, within seed noise of the canonical $0.221$. The Optuna extension corroborates this on mel features: 60 Bayesian-guided trials for both M2 and M3 yield no improvement beyond seed noise. On mel, the data ceiling dominates. On wav2vec2, however, M2 + Optuna does break out of canonical performance ($0.219 \rightarrow 0.251$), suggesting the ceiling is feature-source-specific rather than universal.

## Discussion

### Why Does LR Beat Transformers?

Mel summary statistics capture the average spectral content of a clip — a surprisingly strong signal for context classification. With only 569 training clips from 7 children, the transformer learns speaker identity as effectively as context identity. Mean pooling over the whole clip effectively ignores temporal dynamics, making Model 2 a noisier version of the LR signal.

### When Did Optuna Improve a Model, and When Didn't It?

The discrete 9-point grid for LR ($C \in [10^{-4}, 10^{4}]$, logspaced) was too coarse in the region $[0.001, 0.01]$. The optimal $C = 0.00492$ for mel and $C = 0.01498$ for wav2vec2 lie between grid points — continuous search found them, discrete search could not. These are genuine signal-to-noise ratio improvements.

More surprisingly, **Model 2 on wav2vec2** also benefited from Optuna ($+0.032$ at $|\Delta|/\sigma = 1.6$). This is the only transformer configuration in the project to genuinely beat its canonical baseline. The canonical val-test gap for M2 wav2vec2 was small (comparable to LR), leaving room for the search to add signal rather than overfit the validation child.

For the remaining transformer searches, Optuna found configurations with higher validation F1 but lower or unchanged test F1. The clearest example is M2 mel: the 5-seed rerun gives $0.226 \pm 0.018$, which is $0.023$ below the canonical $0.249 \pm 0.020$ but still within seed noise. The corrected val–test gap is about $0.065$, showing that the search did not transfer reliably from the Shipibo validation child to the Tagalog test child.

This is not simply a hyperparameter problem — it is the language confound. LR MFCC shows a smaller-scale version of the same pattern: inner-CV F1 maximum at $C$=0.01735 (0.246) beat the grid winner (0.243) but transferred to a slightly lower test F1 (0.239 vs. 0.253).

The pattern is consistent across the nine searches: Optuna helps when the validation signal transfers to test, and otherwise gives within-noise changes or degradation.

### The Language Confound

The validation child (child_2) speaks Shipibo-Conibo (22.5% INTER in training). The test child (child_6) speaks Tagalog (46.9% INTER in training). Models tuned on a Shipibo validation set encounter a distributional shift at test time. The Optuna results provide direct evidence: across 60 transformer trials, higher validation F1 does not predict higher test F1 (Figure 8, centre panel). This confirms our interpretation — the confound is structural and cannot be solved by hyperparameter search.

### Speaker Identity is the Dominant Bottleneck

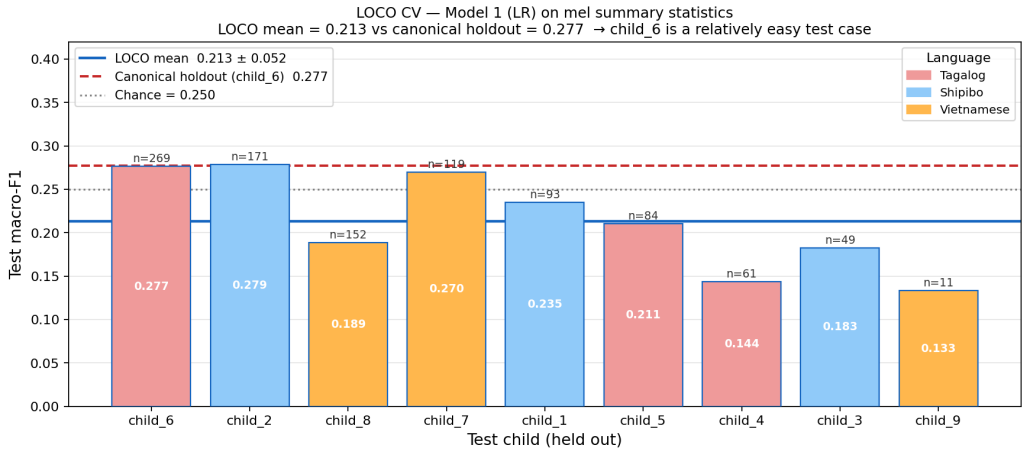



Figure 12: LOCO cross-validation for Model 1 (LR) on mel features. The canonical holdout (child_6) is a relatively easy test case. LOCO mean = $0.213 \pm 0.052$.

The LOCO mean F1 is $0.213 \pm 0.052$, below the fixed child_6 holdout score (0.277 with the canonical grid and 0.311 after Optuna). Per-fold range: 0.133 (child_9) to 0.279 (child_2). Speaker variance ($\sigma \approx 0.052$) dominates LR seed variance, which is zero because the lbfgs solver is deterministic.

### What Attention Concentration Tells Us

On wav2vec2, Model 3's attention collapses to a single frame in 95% of clips — the model uses one contextualised embedding per clip rather than the full sequence. On mel, attention spans ∼17 frames. The learned temperature $\tau < 1.0$ in both configurations indicates the model tried to soften attention; the concentration on wav2vec2 is driven by score magnitudes, not temperature runaway.

## Future Work

**Larger and more diverse participant pool.** The LOCO analysis shows the bottleneck is speaker diversity, not model capacity.

**Matched-language validation strategy.** The language confound suggests future work should control for language distribution when assigning children to train/val/test splits.

**Partial wav2vec2 fine-tuning.** Selectively fine-tuning the top 2–3 transformer layers could better adapt features to child speech.

**LOCO evaluation for all models.** We ran LOCO only for Model 1 due to compute constraints. Running for Models 2 and 3 would reveal whether the language prior affects transformer models more severely.

**Richer context features.** Contextual metadata such as recording setting (home, outdoor, playgroup) could improve classification without exposing participant identity.

## Conclusion

We presented a controlled comparison of three models for classifying child vocalisation contexts across three under-resourced language settings. Logistic regression on mel summary statistics outperforms both transformer architectures — a result robust across feature sources, random seeds, pooling strategies, and LOCO cross-validation. As an extension, Bayesian hyperparameter search (Optuna) yields three genuine gains over canonical: LR mel from $0.277$ to $\mathbf{0.311}$, LR wav2vec2 from $0.263$ to $0.285$, and Model 2 wav2vec2 from $0.219$ to $0.251$. The remaining six (model, feature) searches are within seed noise of canonical or degrade — consistent with the language confound, the $40.9\%$ all-models-wrong rate, and the LOCO-derived speaker-variance estimate. The findings underscore that in small, speaker-limited datasets, model simplicity and careful hyperparameter search outweigh architectural complexity, but that the benefit of careful search depends on whether the validation set is representative of test.

## References

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## Supplementary Tables and Figures

### Per-seed Results

Table 2: Per-seed test macro-F1 for canonical and Optuna-selected runs. LR std = 0.000 (lbfgs is deterministic). Transformer ⋆ rows are 5-seed reruns of Optuna-selected configurations; the M2–Mel row uses the corrected multiseed JSON.

| Model – Feature | s=42 | s=123 | s=456 | s=789 | s=2024 | Mean±Std |
|---|---|---|---|---|---|---|
| LR – MFCC | 0.253 | 0.253 | 0.253 | 0.253 | 0.253 | 0.253±0.000 |
| LR – MFCC ⋆ | 0.239 (deterministic; C=0.01735) | | | | | 0.239±0.000 |
| LR – Mel | 0.277 | 0.277 | 0.277 | 0.277 | 0.277 | 0.277±0.000 |
| LR – Mel ⋆ | **0.311 (deterministic; C=0.00492)** | | | | | **0.311±0.000** |
| LR – wav2vec2 | 0.263 | 0.263 | 0.263 | 0.263 | 0.263 | 0.263±0.000 |
| LR – wav2vec2 ⋆ | **0.285 (deterministic; C=0.01498)** | | | | | **0.285±0.000** |
| M2 – MFCC | 0.242 | 0.216 | 0.267 | 0.219 | 0.168 | 0.222±0.033 |
| M2 – MFCC ⋆ | 0.188 | 0.216 | 0.235 | 0.265 | 0.189 | 0.219±0.029 |
| M2 – Mel | 0.257 | 0.242 | 0.230 | 0.277 | 0.238 | 0.249±0.020 |
| M2 – Mel ⋆ | 0.220 | 0.248 | 0.202 | 0.212 | 0.247 | 0.226±0.018 |
| M2 – wav2vec2 | 0.210 | 0.237 | 0.209 | 0.239 | 0.200 | 0.219±0.018 |
| M2 – wav2vec2 ⋆ | 0.246 | 0.236 | 0.279 | 0.267 | 0.225 | **0.251±0.020** |
| M3 – MFCC | 0.205 | 0.138 | 0.237 | 0.209 | 0.253 | 0.208±0.040 |
| M3 – MFCC ⋆ | 0.244 | 0.248 | 0.180 | 0.163 | 0.251 | 0.217±0.038 |
| M3 – Mel | 0.228 | 0.211 | 0.218 | 0.231 | 0.217 | 0.221±0.013 |
| M3 – Mel ⋆ | 0.246 | 0.246 | 0.239 | 0.210 | 0.222 | 0.233±0.014 |
| M3 – wav2vec2 | 0.250 | 0.262 | 0.241 | 0.263 | 0.214 | 0.246±0.016 |
| M3 – wav2vec2 ⋆ | 0.234 | 0.282 | 0.222 | 0.229 | 0.239 | 0.241±0.021 |

⋆ Optuna-selected configuration. Bold = genuine gain over canonical.

### Optuna Search Parameters

Table 3: Optuna search configuration for each (model, feature) combination. Transformer searches share the same lr/dropout structure. The uploaded M2 mel JSON uses $d \in \{32, 64, 128, 192\}$ and $L \in \{1, 2, 3\}$; M3 adds `dropout_meta`. Feature type is controlled via `config.py` flags.

| Search | Trials | Search space | Objective |
|---|---|---|---|
| LR – MFCC | 80 | $C$ log-uniform $[10^{-6}, 10^{6}]$; seeded with 9-pt grid | Inner LOPO-CV macro-F1 |
| LR – Mel | 80 | ditto | Inner LOPO-CV macro-F1 |
| LR – wav2vec2 | 80 | ditto | Inner LOPO-CV macro-F1 |
| M2 – MFCC | 60 | shared transformer space (3 dropouts) | Val macro-F1 |
| M2 – Mel | 60 | $d \in \{32, 64, 128, 192\}$, $L \in \{1, 2, 3\}$, 3 dropouts | Val macro-F1 |
| M2 – wav2vec2 | 60 | shared transformer space (3 dropouts) | Val macro-F1 |
| M3 – MFCC | 60 | shared + `dropout_meta` | Val macro-F1 |
| M3 – Mel | 60 | shared + `dropout_meta` | Val macro-F1 |
| M3 – wav2vec2 | 60 | shared + `dropout_meta` | Val macro-F1 |

### Ablation Summary

Table 4: Summary of all supplementary ablations and their test F1 outcomes. Top section: feature/architecture ablations. Bottom: full Optuna extension results across all 9 (model, feature) combinations.

| Ablation | Baseline F1 | Change |
|---|---|---|
| *Feature / architecture / metadata ablations:* | | |
| Delta-MFCC features (LR MFCC) | 0.253 | $-0.024$ |
| Seed ensembling (M2 Mel) | 0.249 | $-0.024$ |
| Metadata: age only (M3 wav2vec2) | 0.247 | $+0.001$ |
| Metadata: gender only | 0.221 | $-0.025$ |
| Metadata: none (no conditioning) | 0.233 | $-0.013$ |
| Reg. sweep best (M2 wav2vec2) | 0.219 | $+0.021$ (val only) |
| *Optuna extension (all 9 cells):* | | |
| Optuna LR MFCC ($C$=0.01735) | 0.253 | $-0.014$ |
| Optuna LR Mel ($C$=0.00492) | 0.277 | $+\mathbf{0.034}$ **(genuine)** |
| Optuna LR wav2vec2 ($C$=0.01498) | 0.263 | $+\mathbf{0.022}$ **(genuine)** |
| Optuna M2 MFCC 5-seed | 0.222 | $-0.003$ (within noise) |
| Optuna M2 Mel 5-seed | 0.249 | $-0.023$ (within noise, $\lvert\Delta\rvert/\sigma\approx 1.2$) |
| Optuna M2 wav2vec2 5-seed | 0.219 | $+\mathbf{0.032}$ **(genuine,** $\lvert\Delta\rvert/\sigma=1.6$**)** |
| Optuna M3 MFCC 5-seed | 0.208 | $+0.009$ (within noise) |
| Optuna M3 Mel 5-seed | 0.221 | $+0.012$ (within noise) |
| Optuna M3 wav2vec2 5-seed | 0.246 | $-0.005$ (within noise) |

### LOCO Per-fold Results

Table 5: LOCO cross-validation for Model 1 (LR) on mel features. LOCO mean = $0.213 \pm 0.052$; canonical grid holdout = 0.277 (Optuna holdout = 0.311).

| Child | N clips | Language | Test F1 |
|---|---|---|---|
| child_6 | 269 | Tagalog | 0.277 |
| child_2 | 171 | Shipibo | 0.279 |
| child_7 | 119 | Vietnamese | 0.270 |
| child_1 | 93 | Shipibo | 0.235 |
| child_5 | 84 | Tagalog | 0.211 |
| child_8 | 152 | Vietnamese | 0.189 |
| child_3 | 49 | Shipibo | 0.183 |
| child_4 | 61 | Tagalog | 0.144 |
| child_9 | 11 | Vietnamese | 0.133 |
| **Mean** | | | $\mathbf{0.213 \pm 0.052}$ |